

Domáci úkol 1.1

Nech z je komplexné číslo potom $\operatorname{Re}(z) = \frac{1}{2}(z + \bar{z})$ a $\operatorname{Im}(z) = \frac{1}{2i}(z - \bar{z})$.
Nech $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, $a, b \in \mathbb{C}$ potom

$$\begin{aligned}\langle a\mathbf{u} + b\mathbf{v}, \mathbf{w} \rangle &= \overline{\langle \mathbf{w}, a\mathbf{u} + b\mathbf{v} \rangle} = \overline{a \langle \mathbf{w}, \mathbf{u} \rangle + b \langle \mathbf{w}, \mathbf{v} \rangle} \\ &= \bar{a} \overline{\langle \mathbf{w}, \mathbf{u} \rangle} + \bar{b} \overline{\langle \mathbf{w}, \mathbf{v} \rangle} \\ &= \bar{a} \langle \mathbf{u}, \mathbf{w} \rangle + \bar{b} \langle \mathbf{v}, \mathbf{w} \rangle\end{aligned}$$

Ďalej upravujeme výraz, pomocou vlastností skalárneho súčinu

$$\begin{aligned}\|\mathbf{u} + \mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2 &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle \\ &= \langle \mathbf{u} + \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u} + \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u} - \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u} - \mathbf{v}, \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle + \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle \\ &= 2(\langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{u} \rangle) = 2(\langle \mathbf{u}, \mathbf{v} \rangle + \overline{\langle \mathbf{u}, \mathbf{v} \rangle})\end{aligned}$$

čo sa rovná $4\operatorname{Re}(\langle \mathbf{u}, \mathbf{v} \rangle)$ po predelení oboch strán rovnice číslom 4 dostáveme požadovanú rovnosť.

Podobne

$$\begin{aligned}\|\mathbf{u} - i\mathbf{v}\|^2 - \|\mathbf{u} + i\mathbf{v}\|^2 &= \langle \mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v} \rangle \\ &= \langle \mathbf{u} - i\mathbf{v}, \mathbf{u} \rangle - i \langle \mathbf{u} - i\mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u} + i\mathbf{v}, \mathbf{u} \rangle - i \langle \mathbf{u} + i\mathbf{v}, \mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle + i \langle \mathbf{v}, \mathbf{u} \rangle - i \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{u} \rangle + i \langle \mathbf{v}, \mathbf{u} \rangle - i \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle \\ &= 2i(\langle \mathbf{v}, \mathbf{u} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle) = \frac{2}{i}(\langle \mathbf{u}, \mathbf{v} \rangle - \overline{\langle \mathbf{u}, \mathbf{v} \rangle})\end{aligned}$$

čo sa rovná $4\operatorname{Im}(\langle \mathbf{u}, \mathbf{v} \rangle)$ a požadovanú rovnosť opäť dostaneme predelením oboch strán číslom 4

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